A SUBSPACE MODEL OF SPEECH-WATERMARK SURVIVAL UNDER ANALYSIS–RESYNTHESIS

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ABSTRACT

Speech watermarks are widely reported to fail when audio is regenerated by a vocoder or neural codec. We give this phenomenon a restricted but precise model: a blind launderer is an *analysis–resynthesis channel* $\mathcal{W} = \mathcal{S} \circ \mathcal{A}$. At full generality we prove only what is true: a data-processing inequality (laundering cannot increase the Chernoff detection exponent), and an *exact-erasure condition* — if embedding leaves the analysis output unchanged a.s., the laundered laws coincide and no detector beats chance. In a linear-Gaussian surrogate we obtain closed forms: the channel transmits exactly the equivalence class of the perturbation modulo $\ker \mathcal{A}$, with surviving exponent $\frac{1}{8}\|P\delta\|^2$; a Gaussian *achievable lower bound* R_{LB} (not a capacity) covers blind embedders. The model's empirical content is a measurable *survival predictor*: geometric quantities — each attacker's own analysis sensitivity, or a static magnitude-change fraction. On LibriSpeech (1000 speaker-disjoint test clips, thresholds from 7157 independent calibration clips), across five resynthesis families (spectral inversion, a neural vocoder, and three neural codecs), these predict held-out mel-domain detectability of constructed geometry-probe marks within every attacker (pooled Spearman 0.72 unpaired / 0.45 paired, both perm. $p < 10^{-3}$; replicated across seeds), while SNR and spectral-centroid controls fail. Deployed watermarks (AudioSeal, WavMark, SilentCipher), each at its native transparent strength, lose their calibrated operating point under nearly every lossy channel, in modes we keep separate: *erasure* (separability itself collapses), *calibration failure* (separability remains but the fixed threshold's false-alarm rate explodes on attacked clean audio), and *graceful degradation* (recall falls but detection still works).

Index Terms— audio watermarking, generative AI, provenance, data-processing inequality, hypothesis testing

1. INTRODUCTION

Provenance watermarking embeds a key-dependent signal in speech so a detector can later flag synthetic or traced audio [1, 2, 3]. A consistent empirical finding is fragility to *regeneration*: passing watermarked audio through a vocoder, neural codec, or voice conversion drives deployed detectors toward failure [4, 5]. The community's response is to embed in representations that resynthesis preserves — codec latents (Latent-Mark [6]), speech-feature-aligned marks [7], speaker-latents robust to zero-shot VC [8], and semantic image watermarks before them [9, 10, 11]. The *invariant-alignment idea is thus established practice, not our contribution.*

What the literature lacks is a stated model in which the collapse and its exceptions can be derived. Empirical papers report which schemes die under which attacks; the pattern — AudioSeal survives EnCodec yet fails under mel-inversion; marks with more energy in preserved components survive longer — is left as observation. This paper supplies the missing object: a channel model with exactly-scoped guarantees that *organizes* the deployed-watermark phenomenology (E1), and a *measurable* geometric quantity that, in a controlled setting (constructed marks, a fixed detector), *predicts* post-attack separability out-of-sample (E2). We do not claim the predictor numerically forecasts a given deployed scheme's survival; the two experiments are complementary.

Contributions. (i) A general proposition for blind analysis–resynthesis channels: a Chernoff data-processing inequality, and a sufficient condition for *exact* erasure (§2). Nothing stronger is claimed at this generality. (ii) A linear-Gaussian surrogate where everything is closed-form: the channel transmits the perturbation's class in $\mathbb{R}^n / \ker \mathcal{A}$; mixed perturbations survive *partially* with exponent $\frac{1}{8}\|P\delta\|^2$ (§3). (iii) A Gaussian achievable lower bound R_{LB} for blind embedders — explicitly *not* a watermarking capacity: informed embedding [12, 13] can exceed it and no rate converse is claimed (§3). (iv) The finding that *geometric* quantities — each attacker's own analysis sensitivity $s_{\mathcal{W}}$, or a static magnitude-change fraction — predict *mel-domain* post-attack separability of constructed marks on *held-out, speaker-disjoint* data across five resynthesis families (while budget/centroid controls do not), robust under both paired and unpaired detectability statistics and across seeds (§4). (v) A calibrated operational evaluation of deployed watermarks, separating loss of *separability* (oriented AUC) from loss of the *calibrated operating point* (TPR and *achieved* FPR at a threshold fixed on an independent calibration set) — distinctions routinely conflated (§4).

2. THE ANALYSIS–RESYNTHESIS CHANNEL

A watermark embedder E_κ maps a host $x \sim P_0$ to $E_\kappa(x)$ (additively, $E_\kappa(x) = x + \delta_\kappa$), giving the watermarked law P_1. A detector knowing κ tests $H_0 : P_0$ vs $H_1 : P_1$; with N i.i.d. looks the optimal Bayes error decays as $e^{-N\, C(P_0,P_1)}$, C the Chernoff information [14, 15]. A *blind launderer* is

$$\mathcal{W} \;=\; \mathcal{S} \circ \mathcal{A}, \tag{1}$$

where $\mathcal{A}$ maps a waveform to a representation z (a mel envelope, a codec latent) and $\mathcal{S}$ regenerates a waveform from z with randomness independent of the input. The attacker knows neither κ nor the scheme (adaptive, key-aware removal [16] is out of scope). Imperceptibility is a budget $\rho(\delta; x) \le D$.

Proposition 1 (general channels)**.** *For any channel $\mathcal{W}$,*

$$C(\mathcal{W}_{\#}P_0, \mathcal{W}_{\#}P_1) \;\le\; C(P_0, P_1).$$

Code, proofs, per-clip score arrays, and locked environments: `github.com/appleweiping/resynthesis-watermark-limits`.

If moreover $\mathcal{A}(E_\kappa(x)) = \mathcal{A}(x)$ holds P_0-a.s. and $\mathcal{S}$ depends on its input only through z, then $\mathcal{W}_\# P_1 = \mathcal{W}_\# P_0$; hence $C(\mathcal{W}_\# P_0, \mathcal{W}_\# P_1) = 0$ and no detector beats chance at any sample size.

Proof. The first part is the DPI for the Chernoff family: each $-\log \int p_0^{1-s} p_1^s$ is monotone under Markov kernels and the max over s preserves monotonicity [15]. For the second, $\mathcal{W}(E_\kappa(x)) = \mathcal{S}(\mathcal{A}(E_\kappa(x)), U) = \mathcal{S}(\mathcal{A}(x), U) = \mathcal{W}(x)$ a.s. with $U \perp x$, so the output laws coincide. □

Proposition 1 is all we assert for real nonlinear vocoders and codecs: laundering cannot help the detector, and *exactly* analysis-invariant embeddings are *exactly* erased. It supplies no constants and no prediction of partial survival. Quantitative structure requires a model.

3. LINEAR-GAUSSIAN SURROGATE

Work in whitened coordinates, $P_0 = \mathcal{N}(0, I_n)$, perceptual weighting carried by a masking metric $M \succ 0$ ($\rho(\delta) = \delta^\top M \delta$). Let $\mathcal{A} \in \mathbb{R}^{k \times n}$ have full row rank, $P = \mathcal{A}^+ \mathcal{A}$ the orthogonal projector onto $\mathrm{row}(\mathcal{A})$, and let synthesis resample the complement from the clean prior:

$$\mathcal{W}(x) = Px + (I - P)w, \qquad w \sim \mathcal{N}(0, I_n) \text{ fresh.} \tag{2}$$

Theorem 1 (exact surviving shift and exponent). *Under* (2)*, for a fixed additive shift δ: $\mathcal{W}_\# \mathcal{N}(\delta, I) = \mathcal{N}(P\delta, I)$ and*

$$C(\mathcal{W}_\# P_0, \mathcal{W}_\# P_1) \;=\; \tfrac{1}{8} \|P\delta\|^2. \tag{3}$$

The channel transmits exactly the equivalence class $[\delta] \in \mathbb{R}^n / \ker \mathcal{A}$: the effective perturbation is $P\delta$. Three consequences, stated carefully: $\delta \in \ker \mathcal{A}$ is erased exactly (recovering Prop. 1); *mixed* perturbations survive *partially* (exponent $\frac{1}{8}\|P\delta\|^2 > 0$ whenever $P\delta \neq 0$); and row-space-only perturbations waste no budget on erased components but are *not* the unique surviving set. The budget-optimal exponent is $\max_{\delta^\top M \delta \le D} \frac{1}{8}\|P\delta\|^2 = \frac{D}{8}\, \lambda_{\max}(P, M)$, at the top generalized eigenvector of (P, M).

A rate lower bound (not capacity). Writing surviving shifts in an orthonormal basis of $\mathrm{row}(\mathcal{A})$, a *blind* (host-uninformed, Gaussian codebook) embedder facing a κ-aware but host-blind decoder achieves any rate below

$$R_{\mathrm{LB}} = \max_{Q \succeq 0,\ \mathrm{tr}(M_{\mathrm{row}} Q) \le D} \tfrac{1}{2} \log \det(I + Q) \tag{4}$$

nats per invariant coordinate (host as unit-variance interference; water-filling over M_{row}'s eigenmodes). R_{LB} is an *achievable lower bound for that restricted class*: informed (host-aware) embedding in the Costa/Moulin–O'Sullivan sense [12, 13] can exceed it, and we prove no rate converse. A nullspace embedding has $R_{\mathrm{LB}} = 0$ after $\mathcal{W}$.

From surrogate to real channels. For a real attacker $\mathcal{W}$ the model suggests a *measurable* channel-relative predictor: the attacker's own analysis sensitivity

$$s_{\mathcal{W}}(\delta; x) = \|\mathcal{A}_{\mathcal{W}}(x + \delta) - \mathcal{A}_{\mathcal{W}}(x)\| / \|\delta\|, \tag{5}$$

computed with *that* attacker's $\mathcal{A}_{\mathcal{W}}$ (a mel front-end, a codec encoder). Whether $s_{\mathcal{W}}$ predicts post-attack separability is an *empirical hypothesis* — the surrogate motivates it but does not prove it. Testing it out-of-sample is E2's job; a single mel fraction is never reused across attackers with different $\mathcal{A}_{\mathcal{W}}$.

Table 1. Deployed watermarks under analysis–resynthesis, each at its native transparent strength (*not* PESQ-equalized; per-baseline PESQ/SI-SDR in text). Cells: oriented AUC / TPR / *achieved* FPR at the 1%-FPR threshold fixed on the independent calibration split ($n = 7157$ clean negatives). Outcomes: *erasure* (AUC→0.5, TPR→0); *calibration failure* (AUC retained, achieved FPR explodes on attacked clean audio); *graceful degradation* (AUC and FPR retained, TPR reduced — AudioSeal/DAC).

Channel	AudioSeal	WavMark	SilentCipher
clean	1.00/1.00/0.01	1.00/1.00/0.00	0.99/0.98/0.00
STFT†	0.99/0.90/0.00	1.00/1.00/0.00	0.95/0.90/0.00
mel-inv	0.57/0.01/0.00	0.50/0.00/0.00	0.50/0.00/0.01
Vocos	0.63/0.07/0.01	0.50/0.00/0.00	0.50/0.01/0.00
Vc-EnC	0.52/0.01/0.00	0.50/0.00/0.00	0.50/0.01/0.00
EnC6	0.93/0.99/0.73	0.50/0.00/0.00	0.50/0.00/0.00
EnC3	0.81/0.97/0.80	0.50/0.00/0.00	0.50/0.00/0.01
DAC	0.93/0.64/0.01	0.50/0.00/0.00	0.50/0.01/0.01
SNAC	0.50/0.00/0.00	0.50/0.00/0.00	0.50/0.00/0.01
VC_4	0.51/0.01/0.01	0.50/0.00/0.00	0.50/0.00/0.01
VC_8	0.51/0.02/0.02	0.50/0.00/0.00	0.50/0.01/0.01

4. EXPERIMENTS

Setup. LibriSpeech 16 kHz: test = 1000 speaker-stratified 4 s voice-active clips from test-clean (40 speakers); detector thresholds come *only* from 7157 independent calibration clips (dev-clean, dev-other, and test-other, all speaker-disjoint from the test-clean evaluation set); the predictor's mapping is fit on a dev-clean fit split and evaluated on test-clean only. Ten attackers: STFT-GL (near-lossless control), mel-GL (spectral inversion of THE unified 80-mel analysis), Vocos (neural vocoder), Vocos-EnCodec (learned reconstruction from codec tokens), EnCodec at 6/3 kbps, DAC, SNAC, and kNN-VC self voice-conversion (top-$k \in \{4, 8\}$) — content-, speaker- and prosody-preserving re-synthesis. The *predictor* study (E2) uses the five with a common analysis representation (mel-GL, Vocos, EnCodec, DAC, SNAC). All formal runs fail loudly if any attacker or baseline is missing. Scores, seeds, keys, manifests, and locked environments ship with the repository.

Constructed marks (geometry probes). W.r.t. the unified mel analysis $\mathcal{A}$ we construct, per utterance and key: a *kernel mark* (alternating projections onto quadrature $\cap$ consistent spectrograms — the first-order kernel of magnitude analysis) and a *row mark* (in-phase mel-pattern shifts). Construction quality is *measured*, not assumed: at the operating amplitude the kernel mark's analysis change is only $0.16\times$ that of a random direction of equal norm (residual second-order leakage), whereas the row mark's is $2.32\times$. Their detection uses a genie-aided matched-filter *probe* (the known per-utterance perturbation) — a diagnostic of channel geometry, explicitly not a deployable detector; deployed-detector conclusions come only from the baselines.

E2 — do geometric predictors predict survival? For 108 random directions (kernel/row mixtures, random in-phase) $\times$ randomized budgets, we measure $s_{\mathcal{W}}$ (5) per attacker and post-attack detectability (oriented probe AUC against host variability) in *two detector domains*: waveform correlation and a fixed mel-reading matched filter. Mappings are fit on dev-clean and evaluated on test-clean only; all statistics are within-attacker (rank-pooled), so between-attacker level differences can neither create nor mask an association. Results (Fig. 1): for a fixed mel-reading matched filter, $s_{\mathcal{W}}$ predicts detectability *within every attacker* (per-attacker Spearman $0.80/0.75/0.80/0.77/0.48$; pooled 0.72, permutation $p < 10^{-3}$), as does the static magnitude-change fraction (0.80) — both *geomet-*

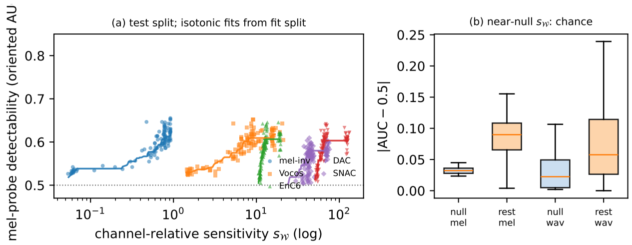

Fig. 1. E2, held-out test split. (a) Channel-relative sensitivity $s_{\mathcal{W}}$ vs. mel-probe detectability, per attacker, with isotonic fits learned on the fit split. (b) For the *unpaired* (operational) detector, directions in the absolute near-null regime stay near chance in both probe domains (a genie paired detector still resolves their second-order leakage; see text).

ric quantities — while budget (SNR, -0.13) and spectral centroid (-0.27) do not. This holds under both the unpaired detectability response above and a stricter *paired*, per-utterance transmission statistic ($s_{\mathcal{W}}$ pooled 0.45, $p < 10^{-3}$; seeds replicate), so it is not an artifact of host variability. Low-$s_{\mathcal{W}}$ directions are *operationally* hard to detect: the 31 directions with $s_{\mathcal{W}} \leq 0.25\times$ median (only on the mel-analysis channels — the codecs' kernels are too small to reach) deviate from chance by median 0.03, maximum 0.04 for the unpaired detector. They are *not* exactly erased, however: our kernel marks are only first-order null (measured second-order leakage $0.16\times$ a random direction's), and a genie *paired* detector still resolves that leakage in the mel domain. So low $s_{\mathcal{W}}$ predicts low operational detectability, not the zero-transmission of the exact-erasure condition. A *waveform*-domain detector shows a weaker, opposite-signed unpaired trend for EnCodec/SNAC ($\rho_s = -0.76/-0.47$), consistent with waveform codecs carrying sub-mel detail; but this does *not* survive the paired transmission statistic ($-0.02/0.10$) or replicate across seeds, so we report it as an observation and do *not* claim a robust detector-domain sign reversal.

E1 — deployed watermarks. AudioSeal [1], WavMark [2], and SilentCipher [3], each at its native full strength — all in the transparent regime but not equalized (clean-embed PESQ $4.42/4.27/4.60$; we do not force a common PESQ, which would push each scheme off its designed operating point, and we note below that AudioSeal carries the most energy). Per (baseline, attacker, key) we report oriented AUC (utterance-clustered bootstrap CIs) and the calibrated operating point — TPR *and achieved FPR* at the 1%-FPR threshold fixed on 7157 independent clean negatives (Clopper–Pearson intervals) — plus an attack-aware recalibration diagnostic (Table 1). We keep two questions separate: does *separability* survive (*raw* AUC, two-sided CI vs. 0.5 — the oriented CI is one-sided by construction) and does the *calibrated operating point* survive (TPR at the fixed-threshold's achieved FPR)? All three baselines are perfect on the near-lossless STFT control. SilentCipher's *continuous* detector falls to chance (raw AUC $\to 0.50$, CI covering 0.5, TPR $\to 0.00$) under all nine lossy channels — a measured separability collapse; WavMark's payload decoder desynchronizes to a constant score under every lossy channel, so its exact 0.50 (zero-width CI) is a deployed-decoder *sync failure*, consistent with (but not itself a measurement of) that collapse. AudioSeal is more varied: separability is *erased* only by SNAC and self-VC (raw AUC 0.50–0.51, CI covers 0.5); under mel-inversion, Vocos and Vocos-EnCodec it retains *weak* separability (raw AUC 0.52–0.63, CI excludes 0.5) but the operating point is dead (TPR ≤ 0.07); under EnCodec it keeps *strong* separability (AUC $0.93/0.81$) yet the fixed threshold's FPR explodes ($0.01 \to 0.73/0.80$) as the codec pushes *clean* audio across the boundary (a *calibration failure*); and under DAC it merely *degrades* (AUC 0.93, FPR 0.01, recall 0.64). Survival is channel-specific in the way the subspace picture suggests — what a scheme's detector reads, relative to what each channel preserves and how it moves clean signals — and no sign/order inversion occurred in the final protocol.

E3 — multi-bit proof of concept (no rate claim). An 8-bit payload in the unified mel domain (frame-differential BPSK, fully *blind* decoder, PESQ 4.2) is carried at BER 0.10 under mel-GL and Vocos and 0.17 under EnCodec and DAC (clean 0.05), but fails under SNAC (BER 0.47); all with binomial CIs. This demonstrates the invariant sub-channel carries bits through most resynthesizers; it is *not* an achievable-rate measurement and we claim no bits/s at target reliability.

5. RELATION TO PRIOR WORK

Attack studies document the collapse [4, 5]; invariant/ latent-aligned embedding is established practice: Latent-Mark optimizes waveforms to shift codec latents with cross-codec transfer [6], Feature-Aligned marks match speech-feature distributions [7], VoiceMark targets VC-invariant speaker latents [8]; in images, semantic watermarks realize the same prescription [9, 10, 11]. Relative to all of these we add what a method paper does not: a stated channel model with exactly-scoped guarantees (Prop. 1, Thm. 1), an explicit account of what is *not* proven (R_{LB} vs capacity [13, 12, 17]), and a falsifiable, channel-relative predictor validated out-of-sample with permutation controls. Our evaluation protocol (independent calibration, oriented-vs-raw separation) addresses measurement pathologies that affect published robustness numbers.

6. SCOPE AND LIMITATIONS

The theory covers blind, non-adaptive laundering; key-aware removal [16] is out of scope. Exact erasure requires exact analysis invariance — our constructed kernel marks realize it only to first order, with measured second-order leakage at operating amplitude. R_{LB} concerns a restricted embedder class; real capacities may be higher. The surrogate's closed forms are not claimed for nonlinear generators — for those we claim only Prop. 1 and the *measured* predictive power of $s_{\mathcal{W}}$. The three deployed baselines run at native strength and are not PESQ-equalized (SI-SDR $26/37/34$ dB); AudioSeal carries the most mark energy, so we claim no cross-scheme ranking; the survival heterogeneity we analyze is *within* AudioSeal (constant energy, varying channel). WavMark's and SilentCipher's collapses are *structural* (a decoder sync failure; a continuous-detector chance collapse), not amplitude effects. Survival statements are about oriented separability or calibrated operating points, never raw sub-0.5 AUC read as erasure.

7. CONCLUSION

A restricted but precise subspace model of analysis–resynthesis explains, with correctly-scoped guarantees, when speech watermarks survive laundering, and yields a measurable geometric predictor that — for a fixed mel-reading detector — generalizes across attackers and held-out speakers under both paired and unpaired detectability statistics. Honest accounting — of what is proved, what is measured, what replicates, and what deployed detectors actually do under attack — is the paper's method as much as its message.

8. REFERENCES

[1] Robin San Roman, Pierre Fernandez, Alexandre Défossez, Teddy Furon, Tuan Tran, and Hady Elsahar, "Proactive detection of voice cloning with localized watermarking," in *International Conference on Machine Learning (ICML)*, 2024.

[2] Guangyu Chen, Yu Wu, Shujie Liu, Tao Liu, Xiaoyong Du, and Furu Wei, "WavMark: Watermarking for audio generation," 2023, arXiv:2308.12770.

[3] Mayank Kumar Singh, Naoya Takahashi, Wei-Hsiang Liao, and Yuki Mitsufuji, "SilentCipher: Deep audio watermarking," in *Interspeech*, 2024.

[4] Patrick O'Reilly, Zeyu Jin, Jiaqi Su, and Bryan Pardo, "Deep audio watermarks are shallow: Limitations of post-hoc watermarking techniques for speech," in *ICLR Workshop on GenAI Watermarking*, 2025.

[5] Yizhu Wen, Ashwin Innuganti, Aaron Bien Ramos, Hanqing Guo, and Qiben Yan, "SoK: How robust is audio watermarking in generative AI models?," 2025, arXiv:2503.19176.

[6] Yen-Shan Chen et al., "Latent-Mark: An audio watermark robust to neural codec compression," 2026, arXiv:2603.05310; to appear, Interspeech.

[7] Haiyun Li et al., "Feature-aligned speech watermarking for robustness to reconstruction distortions," 2026, arXiv:2606.11828.

[8] Haiyun Li, Zhiyong Wu, Xiaofeng Xie, Jingran Xie, Yaoxun Xu, and Hanyang Peng, "VoiceMark: Zero-shot voice cloning-resistant watermarking approach leveraging speaker-specific latents," in *Interspeech*, 2025.

[9] Xuandong Zhao, Kexun Zhang, Zihao Su, Saastha Vasan, Ilya Grishchenko, Christopher Kruegel, Giovanni Vigna, Yu-Xiang Wang, and Lei Li, "Invisible image watermarks are provably removable using generative AI," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[10] Yuxin Wen, John Kirchenbauer, Jonas Geiping, and Tom Goldstein, "Tree-ring watermarks: Fingerprints for diffusion images that are invisible and robust," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[11] Zijin Yang, Kai Zeng, Kejiang Chen, Han Fang, Weiming Zhang, and Nenghai Yu, "Gaussian Shading: Provable performance-lossless image watermarking for diffusion models," in *IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[12] Max Costa, "Writing on dirty paper," *IEEE Transactions on Information Theory*, vol. 29, no. 3, pp. 439–441, 1983.

[13] Pierre Moulin and Joseph A. O'Sullivan, "Information-theoretic analysis of information hiding," *IEEE Transactions on Information Theory*, vol. 49, no. 3, pp. 563–593, 2003.

[14] Herman Chernoff, "A measure of asymptotic efficiency for tests of a hypothesis based on the sum of observations," *The Annals of Mathematical Statistics*, vol. 23, no. 4, pp. 493–507, 1952.

[15] Thomas M. Cover and Joy A. Thomas, *Elements of Information Theory*, Wiley-Interscience, 2nd edition, 2006.

[16] Weikang Ding, Hanqing Guo, Rui Duan, Guangjing Wang, Yuanda Wang, Mingzhe Chen, and Qiben Yan, "Learning to evade: Adaptive attacks on audio watermarking," 2026, arXiv:2606.22310; to appear, Interspeech.

[17] Brian Chen and Gregory W. Wornell, "Quantization index modulation: A class of provably good methods for digital watermarking and information embedding," *IEEE Transactions on Information Theory*, vol. 47, no. 4, pp. 1423–1443, 2001.